

FUPS HHT with UNFAIR COIN

$$P(H) = \frac{2}{3}$$

$$P(T) = \frac{1}{3}$$

$$\begin{cases} E_S = 1 + \frac{1}{3} E_S + \frac{2}{3} E_H \\ E_H = 1 + \frac{1}{3} E_S + \frac{2}{3} E_{HH} \\ E_{HH} = 1 + \frac{2}{3} E_{HH} + \frac{1}{3} E_{HHT} \\ E_{HHT} = 0 \end{cases}$$

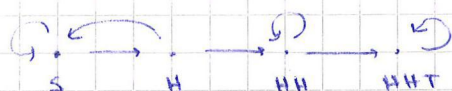
$$E_{HH} = 3$$

$$\begin{aligned} E_S &= 1 + \frac{1}{3} E_S + \frac{2}{3} \left(1 + \frac{1}{3} E_S + 2 \right) \\ &= 1 + \frac{1}{3} E_S + 2 + \frac{2}{9} E_S \end{aligned}$$

$$\left(1 - \frac{1}{3} - \frac{2}{9} \right) E_S = 3$$

$$\frac{9 - 3 - 2}{9} E_S = 3$$

$$E_S = \frac{27}{4}$$



FUPS HTHT - FAIR COIN



ALL PROB = $\frac{1}{2}$

$$S \xrightarrow{1/2} HT \xrightarrow{1/2} HTHT \quad E_S = 20$$

$$\begin{cases} E_S = 1 + \frac{1}{2} E_S + \frac{1}{2} E_H \\ E_H = 1 + \frac{1}{2} E_H + \frac{1}{2} E_{HT} \\ E_{HT} = 1 + \frac{1}{2} E_S + \frac{1}{2} E_{HTH} \\ E_{HTH} = 1 + \frac{1}{2} E_H + \frac{1}{2} E_{HTHT} \\ E_{HTHT} = 0 \end{cases}$$

$$E_{HTH} = 1 + \frac{1}{2} E_H$$

$$\begin{aligned} E_{HT} &= 1 + \frac{1}{2} E_S + \frac{1}{2} \left(1 + \frac{1}{2} E_H \right) \\ &= \frac{3}{2} + \frac{1}{2} E_S + \frac{1}{4} E_H \end{aligned}$$

$$\begin{aligned} E_H &= 1 + \frac{1}{2} E_H + \frac{1}{2} \left(\frac{3}{2} + \frac{1}{2} E_S + \frac{1}{4} E_H \right) \\ &= \frac{7}{4} + \frac{1}{2} E_H + \frac{1}{8} E_H + \frac{1}{4} E_S \end{aligned}$$

$$E_H = \frac{8}{3} \left(\frac{7}{4} + \frac{1}{4} E_S \right) = \frac{14}{3} + \frac{2}{3} E_S$$

$$E_S = 1 + \frac{1}{2} E_S + \frac{1}{2} \left(\frac{14}{3} + \frac{2}{3} E_S \right) = 1 + \frac{1}{2} E_S + \frac{7}{3} + \frac{1}{3} E_S \quad E_S = 20$$

HHTHH

$$S \xrightarrow{2} H \xrightarrow{4} HH \xrightarrow{32} HHTHH$$

$$IE[HHTHH] = 38$$

"SANITY CHECK":



$$\text{All prob} = \frac{1}{2}$$

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH} \\ E_{HH} = 1 + \frac{1}{2}E_{HH} + \frac{1}{2}E_{HHT} \\ E_{HHT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HHHT} \\ E_{HHHT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HHTHH} \\ E_{HHTHH} = 0 \end{cases}$$

E_x = number of flips from x to HHTHH

$$E_{HHTHH} = 1 + \frac{1}{2}E_S$$

$$\begin{aligned} E_{HHHT} &= 1 + \frac{1}{2}E_S + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right) \\ &= \frac{3}{2} + \frac{3}{4}E_S \end{aligned}$$

$$E_{HHT} = 1 + \frac{1}{2}E_{HHHT} + \frac{1}{2}\left(\frac{3}{2} + \frac{3}{4}E_S\right)$$

$$\frac{1}{2}E_{HHT} = 1 + \frac{3}{4} + \frac{3}{8}E_S$$

$$E_{HHT} = 2 + \frac{3}{2} + \frac{3}{4}E_S$$

$$= \frac{7}{2} + \frac{3}{4}E_S$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{7}{2} + \frac{3}{4}E_S\right)$$

$$= 1 + \frac{1}{2}E_S + \frac{7}{4} + \frac{3}{8}E_S$$

$$= \frac{11}{4} + \frac{7}{8}E_S$$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{11}{4} + \frac{7}{8}E_S\right)$$

$$= 1 + \frac{1}{2}E_S + \frac{11}{8} + \frac{7}{16}E_S$$

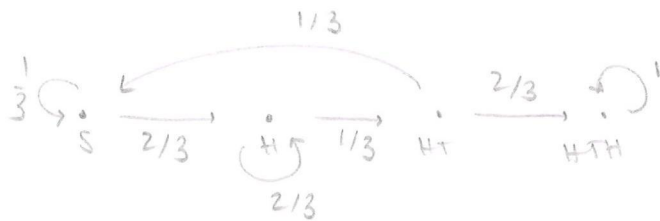
$$\frac{1}{16}E_S = \frac{19}{8}$$

$$E_S = 38 \quad \text{II}$$

UNFAIR COIN

$$IP(H) = \frac{2}{3} \quad IP(T) = \frac{1}{3}$$

What is the expected number of flips until HTH appears for the first time?



E_x = number of flips to go from state x to HTH

$$\begin{cases} E_S = 1 + \frac{1}{3}E_S + \frac{2}{3}E_H \\ E_H = 1 + \frac{1}{3}E_{HT} + \frac{2}{3}E_H \\ E_{HT} = 1 + \frac{2}{3}E_{HTH} + \frac{1}{3}E_S \\ E_{HTH} = 0 \end{cases}$$

$$E_{HT} = 1 + \frac{1}{3}E_S$$

$$E_H = 1 + \frac{1}{3}\left(1 + \frac{1}{3}E_S\right) + \frac{2}{3}E_H$$

$$\frac{1}{3}E_H = 1 + \frac{1}{3} + \frac{1}{9}E_S$$

$$= \frac{4}{3} + \frac{1}{9}E_S$$

$$E_H = 4 + \frac{1}{3}E_S$$

$$E_S = 1 + \frac{1}{3}E_S + \frac{2}{3}\left(4 + \frac{1}{3}E_S\right)$$

$$= 1 + \frac{1}{3}E_S + \frac{8}{3} + \frac{2}{9}E_S$$

$$\left(1 - \frac{1}{3} - \frac{2}{9}\right)E_S = \frac{11}{3}$$

$$\frac{4}{9}E_S = \frac{11}{3} \quad E_S = \frac{11}{3} \cdot \frac{9}{4} = \frac{33}{4}$$

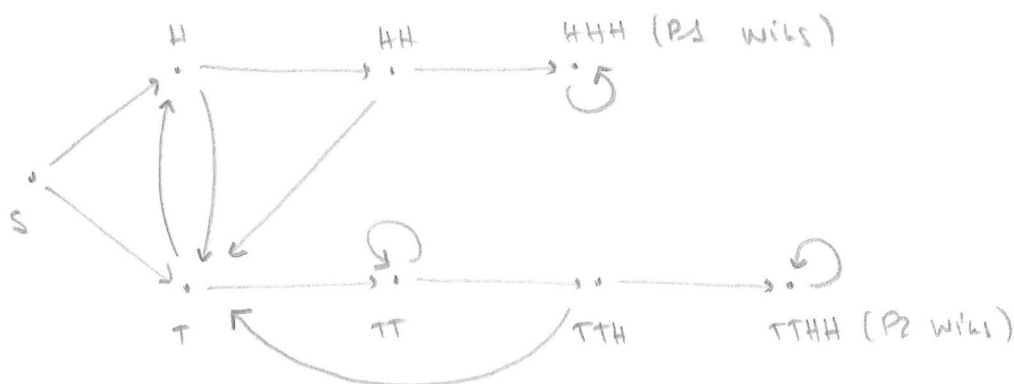
Imagine you have a fair coin, which you toss repeatedly.
Two players observe the sequence of tosses.

P1 waits for HHH

P2 waits for TTH

the one whose pattern appear first is the winner

What is the probability that P1 wins?



P_x = prob of P1 winning

$$P_S = \frac{1}{2} P_H + \frac{1}{2} P_T$$

$$P_H = \frac{1}{2} P_{HH} + \frac{1}{2} P_T$$

$$P_T = \frac{1}{2} P_{TT} + \frac{1}{2} P_H$$

$$P_{HH} = \frac{1}{2} P_T + \frac{1}{2} P_{HHH}$$

$$P_{TT} = \frac{1}{2} P_{TTT} + \frac{1}{2} P_{TTH}$$

$$P_{TTH} = \frac{1}{2} P_T + \frac{1}{2} P_{TTHH}$$

$$P_{HHH} = 1$$

$$P_{TTHH} = 0$$

$$P_{TTHH} = 0$$

$$P_{HHH} = 1$$

$$P_{TTH} = \frac{1}{2} P_T$$

$$P_{TT} = \frac{1}{2} P_{TT} + \frac{1}{2} \left(\frac{1}{2} P_T \right)$$

$$\frac{1}{2} P_{TT} = \frac{1}{4} P_T$$

$$P_{TT} = \frac{1}{2} P_T$$

$$P_{HH} = \frac{1}{2} P_T + \frac{1}{2} P_{HHH} = \frac{1}{2} P_T + \frac{1}{2}$$

$$P_T = \frac{1}{2} P_{TT} + \frac{1}{2} P_H$$

$$= \frac{1}{2} \left(\frac{1}{2} P_T \right) + \frac{1}{2} P_H = \frac{1}{4} P_T + \frac{1}{2} P_H$$

$$\frac{3}{4} P_T = \frac{1}{2} P_H$$

$$P_T = \frac{2}{3} P_H$$

$$P_H = \frac{1}{2} P_{HH} + \frac{1}{2} P_T$$

$$= \frac{1}{2} \left(\frac{1}{2} P_T + \frac{1}{2} \right) + \frac{1}{2} P_T$$

$$= \frac{3}{4} P_T + \frac{1}{4} = \frac{3}{4} \cdot \frac{2}{3} P_H + \frac{1}{4}$$

$$\frac{1}{2} P_H = \frac{1}{4} \quad P_H = \frac{1}{2}$$

$$P_S = \frac{1}{2} P_H + \frac{1}{2} P_T$$

$$= \frac{1}{2} P_H + \frac{1}{3} P_H$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$